

Paradigm Justification Statement

Research Title: Digital Transformation and Its Relationship with the Quality and Efficiency of Higher Education: A Case Study of the Universidad Nacional Mayor de San Marcos

UNMSM – Doctoral Program in Deep Technologies

Course: Research Methods & Scientific Integrity in AI

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1. Topic and Context

Digital transformation has evolved from being a technological initiative into a strategic capability that shapes the competitiveness, quality, and sustainability of higher education institutions. Universities are expected not only to digitize administrative processes, but also to redesign teaching, research, governance, and student services through digital innovation. Consequently, educational quality and institutional efficiency increasingly depend on the effective integration of digital technologies, organizational change, and data-informed decision-making.

In Peru, public universities face the challenge of responding to rapid technological change while ensuring academic excellence, transparency, and efficient resource management. The Universidad Nacional Mayor de San Marcos (UNMSM), as the oldest university in the Americas and one of the country's leading research institutions, has implemented multiple digital initiatives to modernize its academic and administrative functions. Despite these advances, there is limited empirical evidence explaining how digital transformation maturity is associated with educational quality and institutional efficiency across organizational units. This research addresses this gap by generating evidence that supports strategic decision-making and contributes to the modernization of public higher education.

2. Preliminary Research Question

Primary Research Question

How is digital transformation maturity associated with the quality and efficiency of higher education across organizational units at the Universidad Nacional Mayor de San Marcos?

Supporting Questions

- Which dimensions of digital transformation maturity are associated with the quality of higher education across organizational units at the Universidad Nacional Mayor de San Marcos?
- How is digital transformation maturity associated with the efficiency of academic and administrative processes across organizational units at the Universidad Nacional Mayor de San Marcos?
- What organizational, technological, and human factors facilitate or hinder the successful implementation of digital transformation at the Universidad Nacional Mayor de San Marcos?
- What strategic recommendations can be proposed to strengthen digital transformation and support improvements in the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos?

3. Chosen Paradigm and Justification

This study adopts the pragmatic paradigm because its primary objective is to understand a real institutional problem while generating actionable knowledge for decision-makers. Digital transformation is a complex socio-technical phenomenon involving technological infrastructure, organizational culture, leadership, governance, teaching innovation, and human behavior. No single methodological tradition is sufficient to capture all these dimensions.

The pragmatic paradigm prioritizes the research question over methodological preferences, allowing the integration of quantitative evidence (institutional indicators, surveys, statistical models) with qualitative evidence (interviews, policy documents, and stakeholder perspectives). This integration enables a comprehensive explanation of the relationships among digital transformation maturity, educational quality, and institutional efficiency while

producing findings that are academically rigorous and practically relevant. Therefore, pragmatism provides the most appropriate philosophical foundation for examining digital transformation as a measurable organizational process within the institutional context of UNMSM.

4. Implications of the Chosen Paradigm

The selected paradigm leads to an explanatory and comparative mixed-methods case study design. Quantitative analyses will assess associations and differences in measurable indicators of educational quality and institutional efficiency across organizational units with naturally occurring levels of digital transformation maturity, whereas qualitative evidence will help explain the organizational, technological, and human factors underlying the observed relationships. The study is expected to generate evidence-based recommendations for university leaders, policymakers, and higher education institutions seeking to strengthen digital governance, educational quality, and institutional efficiency.

5. One Doubt or Tension

A key methodological tension lies in distinguishing the relationship between digital transformation maturity and educational quality and institutional efficiency from the influence of other variables, such as leadership, funding, regulatory changes, faculty capabilities, and organizational culture. Because organizational units are not randomly assigned to different levels of digital transformation maturity, selection and confounding may affect the observed associations. In addition, reverse causality is possible: better-performing units may have greater capacity to adopt digital technologies. Therefore, the study will avoid interpreting observed associations as definitive causal effects and will rely on rigorous variable operationalization, methodological triangulation, quantitative controls, and qualitative evidence to strengthen the validity and credibility of its conclusions.

Concluding Statement

This research is expected to contribute not only to the academic understanding of digital transformation in higher education but also to the

development of evidence-based strategies capable of enhancing the quality, efficiency, and long-term sustainability of public universities in Peru.